

David William Erwin

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EDUCATION

Czech Technical University (ČVUT)

August 2022 – June 2025

Bachelor of Science in Electrical Engineering and Computer Science

Prague, Czech Republic

- **Machine Learning Research Thesis — CERN ATLAS**
- Developed a transformer-based classifier ($\approx 1.2\text{M}$ parameters) for rare Higgs boson signals in noisy high-energy physics datasets from CERN ATLAS, achieving an AUC of 0.804 and F1-score of 0.705. Engineered custom preprocessing pipeline with particle-specific feature scaling, applied advanced regularization techniques, and improved interpretability via self-attention visualization aligned with physical observables.
- **Relevant Courses:** Statistical Machine Learning, Pattern Recognition, Computer Architectures, Cybernetics, C Programming, Logic Systems and Processors, Quantum Computing, Probability and Statistics.

EXPERIENCE

Software Engineer - Data Pipeline

September 2025 – Present

Mastercard

Prague, Czech Republic

- Redesigned e-commerce adjustment step in SpendingPulse core pipeline with modern idempotent design, significantly improving execution speed and maintainability.
- Co-led SpendingPulse Near-Real-Time pipeline migration to Databricks; cut operational expenses and boosted developer experience via modern Spark/Delta Lake workflows.
- Co-led a 2-person team consolidation of two legacy tourism pipelines into a unified hybrid pipeline; preserved complex geography-level logic while modernizing data flows and eliminating technical debt.
- Major contributor to dev tools including MCP servers and auto-generation of components for new pipeline features.

PROJECTS

drillbit

February 2024 – Present

Expo, OpenRouter

- Built an interview prep app that adapts to a user-defined focus area through context-engineered prompts.
- Designed an elegant AI coach that guides users through problems with minimal interruption to the solving flow.
- Implemented modern text streaming and low-friction UI/UX for fast, unobtrusive AI-assisted practice.

Polymancer

December 2024 – Present

Expo, Supabase, Upstash Redis, Cloudflare Workers, OpenRouter

- Built a trading Polymarket agent that lets users manage and monitor a custom strategy 24/7 from mobile.
- Designed a hold-first agent orchestrator that turns market, position, and signal data into trade decisions.
- Implemented shared risk controls and real-time copy, snipe, and arbitrage modules within one agent workflow.

Swordfish

July 2024 – Present

Next.js, Supabase, Redis, Docker, massive.com, TradingView

- Built a real-time futures analytics terminal centered on fast, chart-driven market analysis.
- Designed a Bun market-data hub with Redis hot caching to power live updates and responsive views.
- Implemented multi-symbol comparison, weighted spreads, and front-month contract handling for futures analysis.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Kotlin, C

Full-Stack: React, Expo, Next.js, Bun, Zod, Zustand, Tailwind CSS, Uniwind, Redis, Postgres

Cloud & DevOps: AWS, Terraform, Cloudflare Workers, Docker, GitHub Actions, Tailscale, Fly.io, Railway

Data / ML Tools: Apache Spark, Databricks, PyTorch, Optuna, Pandas

CERTIFICATES

Databricks Certified Data Engineer Associate | *Databricks Cloud*

October 2024

- Validated expertise in building ETL pipelines and managing large datasets with Apache Spark on Databricks.

Machine Learning Specialization | *Stanford Online - Andrew Ng*

September 2024

- Implemented supervised/unsupervised learning, backpropagation, and gradient descent algorithms from scratch.